

Uniform Trace-Class Möbius–Bergman Families with Quantified Order Sensitivity

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Abstract

We promote a two-branch Möbius transfer operator from one digit pair to the rectangle $(a, b) \in [3, 7/2] \times [6, 7]$. The closed branch images remain separated by at least $1/45$, while the unweighted composition sum on normalized Bergman space is trace class with norm at most $89/16$ and is trace-norm Lipschitz in both parameters. Every word has an exact quadratic-surds trace, giving all power traces and a primitive Fredholm product. The non-cyclic same-count words $aaabb$ and $aabab$ have matrix trace gap $a(b - a)^2 \geq 175/8$, so order sensitivity holds uniformly. These are source-dynamical statements, not an arithmetic or target spectral identification.

1 Family-level progress

Let $\mathbb{D} = \{z : |z| < 1\}$ and equip $A^2(\mathbb{D})$ with normalized area, so $e_n(z) = \sqrt{n+1}z^n$ is orthonormal. Freeze

$$\phi_x(z) = \frac{1}{x+z}, \quad \mathcal{L}_{a,b} = C_{\phi_a} + C_{\phi_b}, \quad (a, b) \in \mathcal{R}_* = [3, 7/2] \times [6, 7].$$

The advance over a fixed pair is uniform: no conclusion below depends on choosing one parameter sample.

2 Uniform geometry and nuclearity

The image of $\overline{\mathbb{D}}$ under ϕ_x is the closed disk with center $x/(x^2 - 1)$ and radius $1/(x^2 - 1)$. Its real extremes are $1/(x+1)$ and $1/(x-1)$, hence the gap between the two images is

$$g(a, b) = \frac{1}{a+1} - \frac{1}{b-1} \geq \frac{1}{9/2} - \frac{1}{5} = \frac{1}{45}.$$

This is a rectangle theorem rather than a corner sample: $1/(a+1)$ decreases with a , whereas $-1/(b-1)$ increases with b . The minimum is therefore attained at the exact corner $(a, b) = (7/2, 6)$.

The endpoint is substantive. On the initially tempting larger rectangle $[3, 4] \times [6, 7]$, the same monotonic calculation gives $g(4, 6) = 1/5 - 1/5 = 0$. The two open images remain disjoint at that corner, but their closures are tangent. Thus pointwise open-image separation cannot be promoted to a positive uniform closed gap on the larger rectangle.

Theorem 1 (uniform trace class). *For every $(a, b) \in \mathcal{R}_*$, $\mathcal{L}_{a,b}$ is trace class and $\|\mathcal{L}_{a,b}\|_1 \leq 89/16$.*

Proof. The rank-one expansion $C_{\phi_x} f = \sum_{n \geq 0} \langle f, e_n \rangle \sqrt{n+1} \phi_x^n$ is nuclear because $\sup_{\mathbb{D}} |\phi_a| \leq 1/2$ and $\sup_{\mathbb{D}} |\phi_b| \leq 1/5$. Using $\sqrt{n+1} \leq n+1$ gives

$$\|\mathcal{L}_{a,b}\|_1 \leq \sum_{n \geq 0} (n+1)2^{-n} + \sum_{n \geq 0} (n+1)5^{-n} = 4 + \frac{25}{16}.$$

□

Proposition 1 (trace-norm continuity). *For two parameters in $\mathcal{R}_*$,*

$$\|\mathcal{L}_{a,b} - \mathcal{L}_{a',b'}\|_1 \leq 4|a - a'| + \frac{5}{32}|b - b'|.$$

Proof. The resolvent identity gives $|\phi_x - \phi_y| \leq |x - y|/(m - 1)^2$ on $\mathbb{D}$ when $x, y \geq m$. If both symbols have modulus at most r , then $|\phi_x^n - \phi_y^n| \leq nr^{n-1}|\phi_x - \phi_y|$. The same rank-one decomposition as in Theorem 1 now bounds the trace norm by

$$\frac{|x - y|}{(m - 1)^2} \sum_{n \geq 1} (n + 1)nr^{n-1} = \frac{2|x - y|}{(m - 1)^2(1 - r)^3}.$$

The choices $(m, r) = (3, 1/2)$ and $(6, 1/5)$ give 4 and $5/32$. $\square$

3 All words and the determinant

Put $M_x = \begin{pmatrix} 0 & 1 \\ 1 & x \end{pmatrix}$. For $M_w = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, write $t = \text{tr } M_w$, $\delta = (-1)^{|w|}$, and $\Delta = t^2 - 4\delta$. Directly solving the fixed equation gives

$$z_w = \frac{A - D + \sqrt{\Delta}}{2C}, \quad \lambda_w = \frac{t - \sqrt{\Delta}}{t + \sqrt{\Delta}}, \quad \text{Tr } C_{\Phi_w} = \frac{1}{2} + \frac{t}{2\sqrt{\Delta}}.$$

Here $M_{x_1} \cdots M_{x_n}$ represents $\phi_{x_1} \circ \cdots \circ \phi_{x_n}$. Composition operators multiply in the reversed function order. Reversal is a bijection on all length- n words, so it preserves the complete trace sum without asserting a false termwise order equality. Expansion of $\mathcal{L}_{a,b}^n$ therefore gives every power trace. Regrouping repetitions in the Fredholm logarithm yields

$$\det(I - z\mathcal{L}_{a,b}) = \prod_{[p]} \prod_{k \geq 0} (1 - z^{|p|} \lambda_p^k), \quad |z| < \frac{1}{2}. \quad (1)$$

Cyclic rotation preserves the trace and determinant of a matrix product, so λ_p is well defined on the class $[p]$. There are at most $2^\ell/\ell$ primitive classes of length ℓ , while the chain rule gives $|\lambda_p| \leq 4^{-\ell}$. Consequently the absolute factor arguments are bounded by a constant multiple of $\sum_{\ell \geq 1} (2|z|)^\ell/\ell$. This proves raw-product convergence for $|z| < 1/2$; the near-zero logarithmic identity extends across that disk by the identity theorem. The left side is entire by Theorem 1, but analytic continuation of the determinant does not claim convergence of the displayed raw factors outside their proved disk.

4 Uniform order sensitivity

The words $aaabb$ and $aabab$ have the same symbol population and are not cyclic rotations. Exact multiplication gives

$$\begin{aligned} t_1 &= a^3b^2 + a^3 + 2a^2b + 2ab^2 + 3a + 2b, \\ t_2 &= a^3b^2 + 4a^2b + ab^2 + 3a + 2b, \end{aligned}$$

so

$$t_1 - t_2 = a(b - a)^2 \geq \frac{175}{8}.$$

Every coefficient in the displayed polynomial for t_1 is positive on $\mathcal{R}_*$, so it is coordinatewise increasing and $t_1 \leq t_1(7/2, 7) = 10731/4$. For $F(t) = 1/2 + t/(2\sqrt{t^2 + 4})$, $F'(t) = 2/(t^2 + 4)^{3/2}$. The mean-value theorem therefore gives

$$F(t_1) - F(t_2) \geq \frac{2800}{(10731^2 + 64)^{3/2}} > 0.$$

Both traces are positive throughout $\mathcal{R}_*$, so the derivative bound is applied on exactly the interval between t_2 and t_1 ; no squaring or branch choice is hidden in this comparison.

5 Exact receipt and boundary

At nine rational sentinels, the release reconstructs all words through period ten: $9(2^1 + \dots + 2^{10}) = 18,414$ rooted receipts and $9 \cdot 226 = 2,034$ primitive-parameter receipts. A separate symbolic run performs 18,379 exact checks; replay is byte-identical and 40 repaired-hash plus one stale-hash mutations are rejected. The grid audits the implementation and is not the proof of uniformity.

The exact verdict is

(A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL).

There is no target divisor, prime or zero table, arithmetic Euler factor, root number, automorphy claim, unitary lift, or Hilbert–Pólya operator. Route B is unauthorized under NO_BAD_EULER_OR_ROOT_NUMBER.

6 Conclusion

C137 proves that nonlinear order sensitivity and trace-class determinant control persist on a compact parameter family. Connecting this source-owned family to any external target is a separate unsupported problem.